STREAMLINING TICKET ASSIGNMENT FOR EFFICIENT SUPPORT OPERATIONS

Full Project Report

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1. INTRODUCTION

1.1 Project Overview

In modern organizations, support systems play a crucial role in maintaining customer satisfaction and operational efficiency. Traditional ticket management systems often rely on manual assignment, which leads to delays, misrouting, and inefficiencies.

This project focuses on developing an **automated ticket assignment system** that intelligently routes tickets to the appropriate teams based on predefined rules, roles, and workflows. The system ensures faster resolution, improved accuracy, and enhanced productivity.

1.2 Purpose

The purpose of this project is to:

- Eliminate manual ticket assignment
- Reduce delays and errors
- Improve efficiency of support teams
- Enhance user/customer satisfaction
- Provide a scalable and reliable solution

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2. IDEATION PHASE

2.1 Problem Statement

Manual ticket assignment in support systems leads to:

- Delays in issue resolution

- Incorrect routing of tickets
- Increased workload for support teams
- Poor customer experience

2.2 Empathy Map Canvas

User Type: Support Team Member

- **Says:** “Too many tickets to handle manually”
- **Thinks:** “Automation would reduce workload”
- **Does:** Assigns and reassigns tickets
- **Feels:** Stressed and overwhelmed

User Type: Customer

- **Says:** “My issue is urgent”
- **Thinks:** “Why is it taking so long?”
- **Does:** Raises tickets and follows up
- **Feels:** Frustrated and dissatisfied

2.3 Brainstorming

Generated ideas:

- Automated ticket routing
- Role-based access control
- Workflow automation
- Notification system
- Dashboard tracking

Selected Idea: Automated ticket assignment using workflows and roles

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3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. User logs into system
2. Creates a support ticket
3. System processes ticket

4. Ticket is automatically assigned
5. Support team resolves issue
6. User receives updates

3.2 Solution Requirements

Functional Requirements

- User registration & login
- Ticket creation & tracking
- Automated assignment
- Role & group management
- Notifications

Non-Functional Requirements

- Security (authentication & RBAC)
- Performance (fast processing)
- Scalability (handle large users)
- Reliability (accurate routing)

3.3 Data Flow Diagram

Level-0:
User → System → Support Team → User

Level-1:

- Ticket Creation
- Classification
- Routing Engine
- Ticket Handling
- Status Updates

3.4 Technology Stack

- **Frontend:** React.js
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Platform:** ServiceNow (for workflow simulation)

- **Tools:** Postman, GitHub

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4. PROJECT DESIGN

4.1 Problem-Solution Fit

The solution directly addresses inefficiencies in manual ticket assignment by automating routing, reducing delays, and improving accuracy.

4.2 Proposed Solution

A system that:

- Automatically assigns tickets
- Uses role-based access control
- Provides dashboards and notifications
- Ensures efficient workflow management

4.3 Solution Architecture

Three-Tier Architecture:

1. **Presentation Layer**
    - User Interface (React)
2. **Application Layer**
    - Business logic (Node.js)
    - Routing engine
    - Workflow automation
3. **Data Layer**
    - MongoDB database

Flow:
User → UI → Backend → Database → Routing Engine → Support Team → User

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5. PROJECT PLANNING & SCHEDULING

5.1 Agile Methodology

- Project divided into 4 sprints
- Each sprint includes development & testing
- Story points used for estimation

Sprint Summary

Sprint	Focus Area
Sprint-1	Registration, Login, Ticket Creation
Sprint-2	Ticket Assignment, Roles, Groups
Sprint-3	Workflow Automation
Sprint-4	Security, Testing

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6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

- Tested all modules:
    - Login
    - Ticket creation
    - Assignment
    - Dashboard

6.2 Performance Testing

- Verified system handles multiple users
- Tested response time
- Ensured quick ticket routing

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7. RESULTS

7.1 Output

- Successful ticket creation
- Automatic assignment to correct team
- Real-time status updates
- Dashboard monitoring

(Attach screenshots in final report)

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8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces manual effort
- Improves efficiency
- Faster issue resolution
- Better user satisfaction

Disadvantages

- Initial configuration complexity
- Requires proper rule setup

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9. CONCLUSION

The project successfully demonstrates an automated ticket assignment system that enhances support operations. By integrating workflow automation and role-based access control, the system improves efficiency, reduces errors, and ensures faster resolution of issues.

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10. FUTURE SCOPE

- AI-based ticket classification
- Chatbot integration
- Advanced analytics dashboard
- Mobile application support

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11. APPENDIX

- **Source Code:** (Add GitHub link)
- **Dataset:** Not applicable
- **GitHub Link:** (Add link)
- **Demo Link:** (Add video/demo link)

FINAL NOTE

This is a **complete, high-quality report** suitable for:

- College submission
- Viva
- PDF/Word upload